

Risk Neutral Valuation

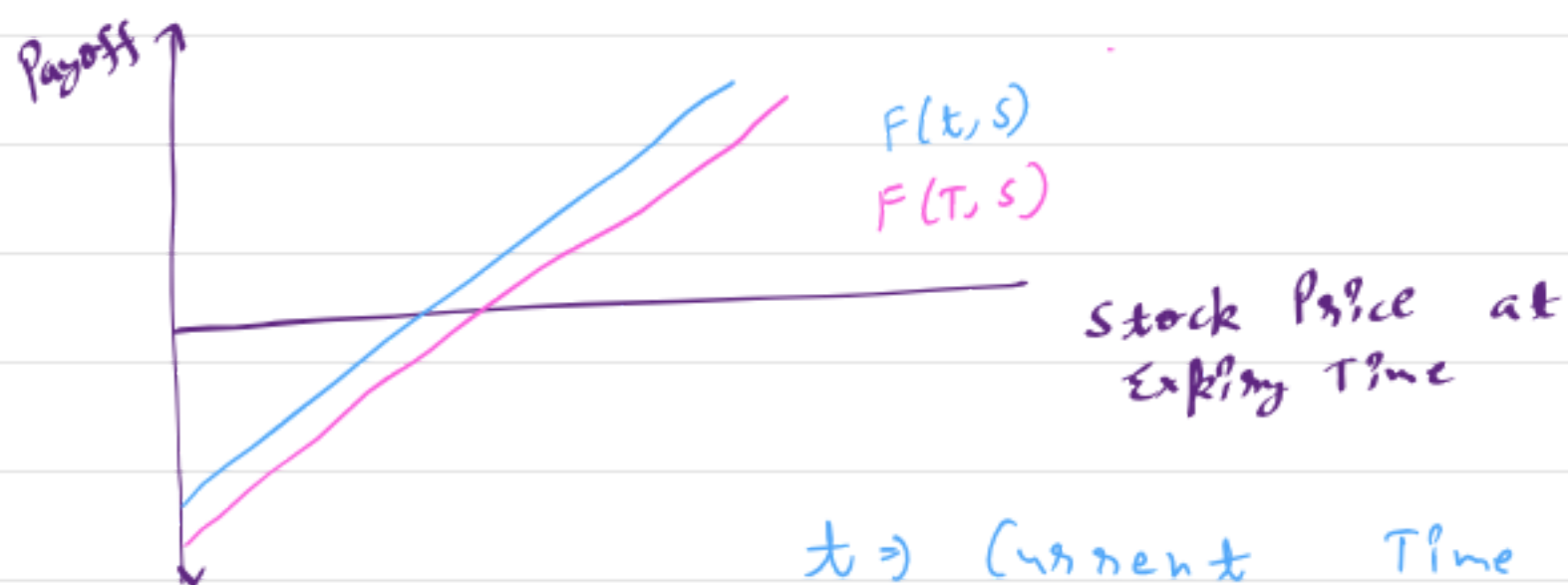
⇒ Expected value is zero

Forward Contract

We have to buy the given stock at time of expiration at Strike Price "K".

$$\text{Payoff} = S_T - K$$

S_T ⇒ Final Price of stock

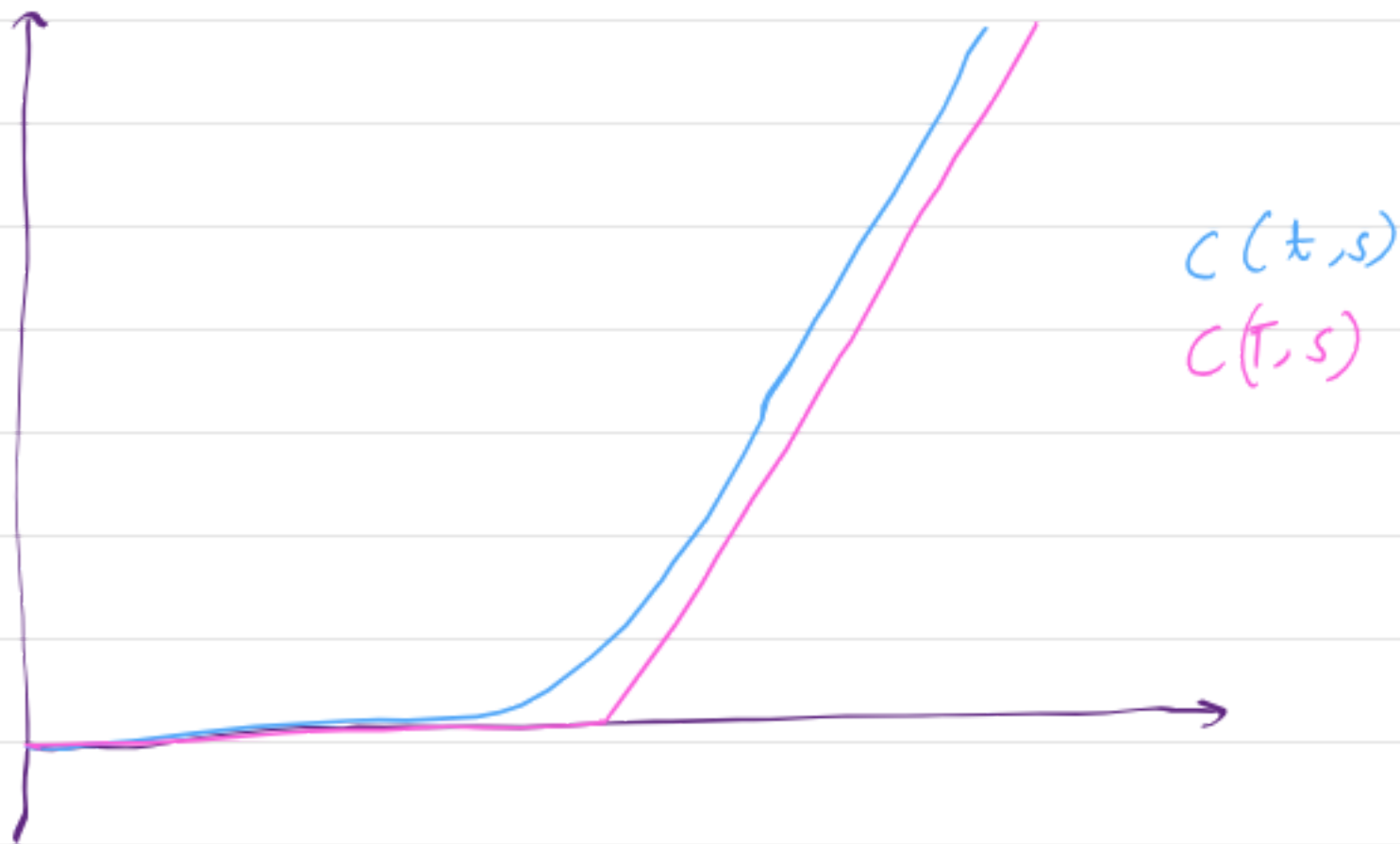


⇒ Blue line is above Pink because money today has more worth than money later.

European Call option

We buy the "right" to buy the option for strike Price (No obligation)

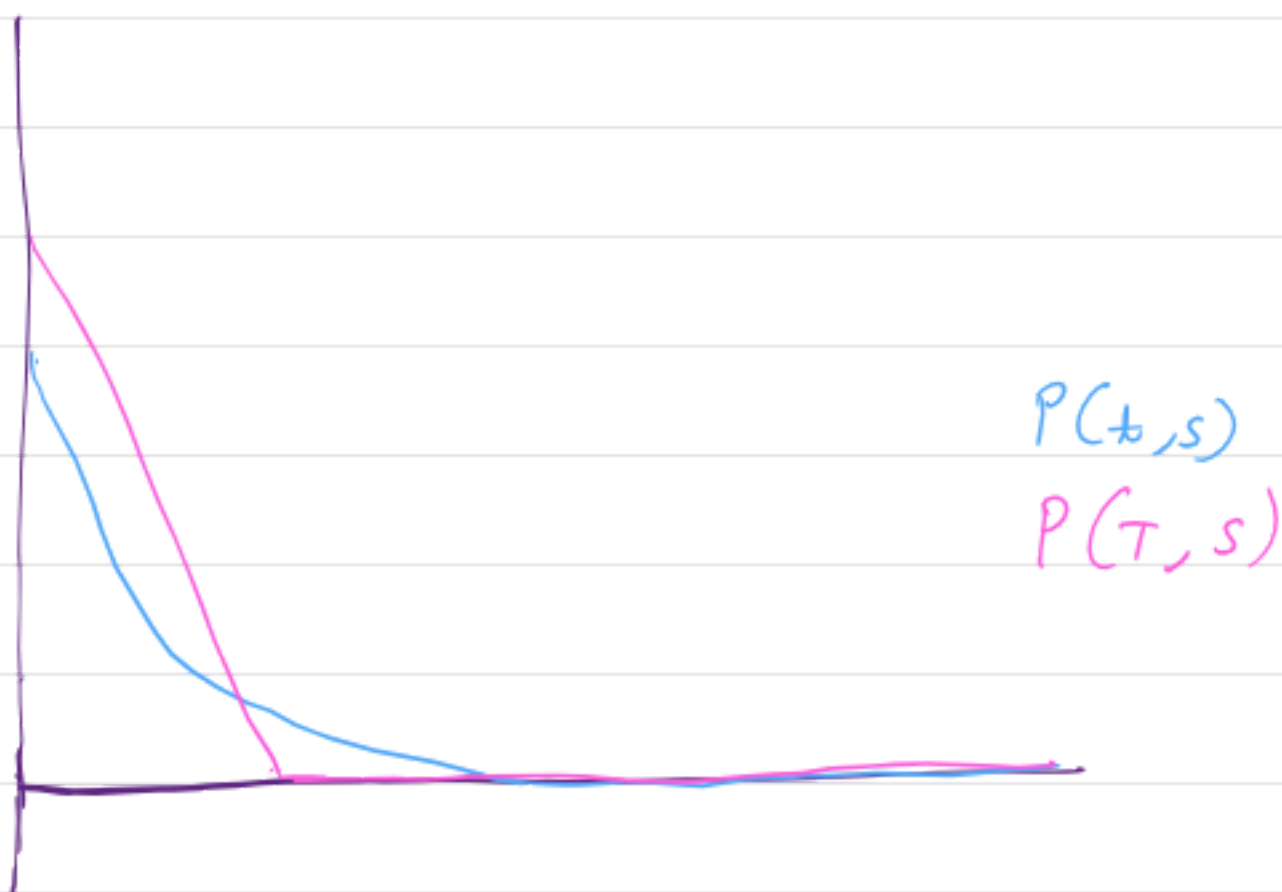
Payoff $\Rightarrow \max(s_T - K, 0)$ at time T .



$\Rightarrow$ Blue line is curved because of increment of max function at time " t ".

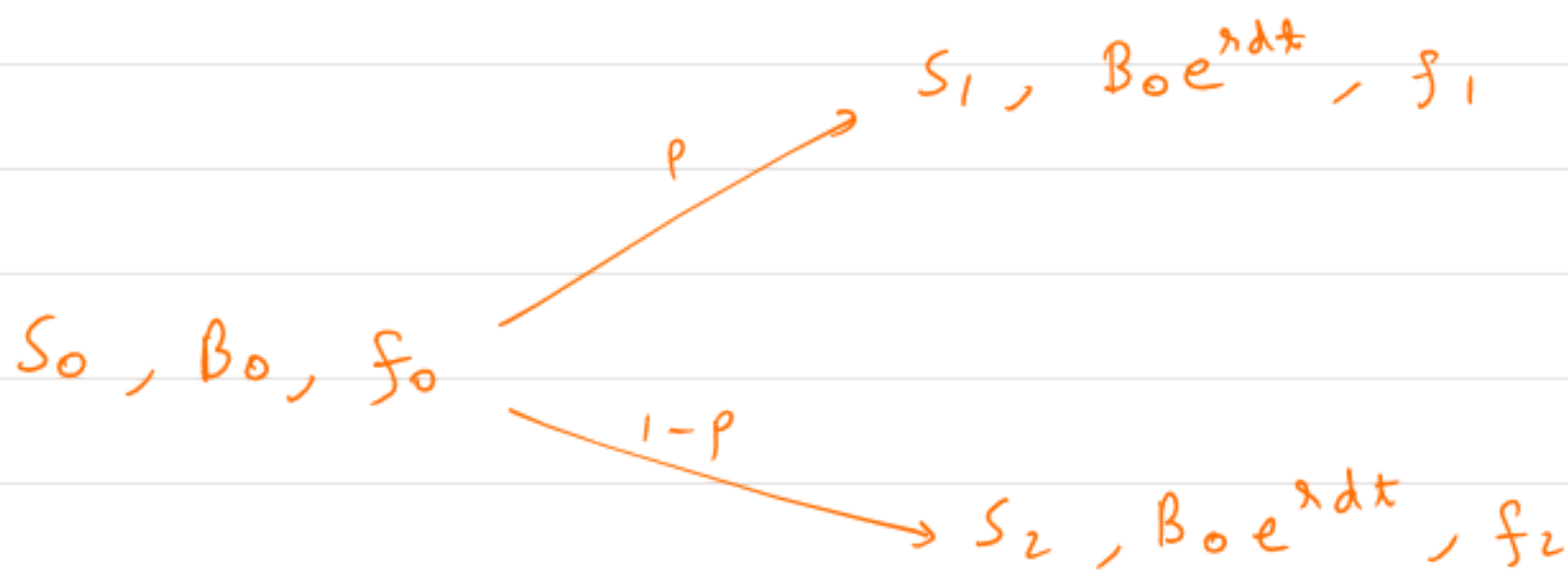
European Put option

$\Rightarrow$ Similar but "Right" to sell



- 1) Given current price of the stock and assumptions on the dynamic of stock price, there is no uncertainty about the price of derivative.
- 2) The price is defined only by the price of the stock & not by the risk preferences of the market participants.
- 3) Mathematical apparatus allows to compute current price of a derivative and its risks, given certain assumptions about the market.

One Step Binomial Tree



$B_0 \rightarrow$ Risk Free Bank Account

$S \rightarrow$ Stock

$r \rightarrow$ Interest Rate

$f \rightarrow$ Derivative claim

Consider Forward Contract

$$\underbrace{p(S_1 - K) + (1-p)(S_2 - K)}_{\text{Ind. Payoff}} = \underbrace{pS_1 + (1-p)S_2}_{\text{Collective Payoff}} - K$$

$$K_0 = pS_1 + (1-p)S_2$$

*** If $p = 1/2$, $K_0 = \frac{(S_1 + S_2)}{2}$

Replicating Portfolio

1. Borrow \$ S_0 to buy a stock. Enter Forward contract with strike K_0
2. In time dt deliver stock in exchange for K_0 and repay \$ $S_0 e^{rdt}$

If $S_0 e^{rdt} > K_0$ Always Profit
 $S_0 e^{rdt} < K_0$ Always Loss

So,

$$S_0 e^{rdt} = K_0 \quad * * *$$

One step Binomial (cont.)

Find a, b s.t.

$$f_1 = a S_1 + b B_0 e^{rdt}$$

$$f_2 = a S_2 + b B_0 e^{rdt}$$

$$f_0 = a S_0 + b B_0$$

So,

$$a = \frac{f_1 - f_2}{S_1 - S_2}$$

$$b = \frac{S_1 f_2 - S_2 f_1}{(S_1 - S_2) B_0 e^{rdt}}$$

$$f_0 = e^{-rdt} \left(S_0 e^{rdt} \frac{f_1 - f_2}{S_1 - S_2} + \frac{S_1 f_2 - S_2 f_1}{S_1 - S_2} \right)$$

$$f_0 = e^{-rdt} (f_1 q + f_2 (1-q))$$

$$q = \frac{(S_0 e^{rdt} - S_2)}{S_1 - S_2} \quad 0 < q < 1$$

$$S_1 q + S_2 (1-q) = S_0 e^{rdt}$$

Black Scholes Eqⁿ

$$dS = \mu S dt + \sigma S dW$$

$S \rightarrow$ log normal

$W \sim N(0, t)$ (Brownian Motion)

$\Rightarrow$ We want replicating portfolio s.t.,

$$df = a dS + b dB$$

Using Ito's Lemma

$$df(s, t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial s} ds + \frac{\partial^2 f}{\partial s^2} (ds)^2$$

$$(ds)^2 = \sigma^2 s^2 dt \quad (\text{Quadratic Variance})$$

$$df = a dS + b dB$$

$$dB = r B dt$$

$$dS = \mu S dt + \sigma S dW$$

$$\begin{aligned} df &= a(\mu S dt + \sigma S dW) + b r B dt \\ &= (a \mu S + b r B) dt + a \sigma S dW \end{aligned}$$

LHS $\Rightarrow$

$$\left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} \mu S + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) dt + \frac{\partial f}{\partial S} \sigma S dW$$

Comparing terms,

$$a = \frac{\partial f}{\partial S}, \quad b \text{ or } B = \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2$$

$bB = f - as$ is deterministic (known in adv.)

$$dB = rB dt$$

$$B = \frac{f - as}{b}$$

$$d \left(\frac{f - as}{b} \right) = r \frac{f - as}{b} dt$$

Substituting $df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S} dS + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 dt$

and $a = \frac{\partial f}{\partial S}$

$$\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 + \frac{\partial f}{\partial S} rS - rf = 0$$

Black Scholes $E_q^{n^{***}}$

$\Rightarrow$ Any tradable derivative satisfies the eqⁿ

$\Rightarrow$ There is no dependence on actual drift μ

$\Rightarrow$ We have a hedging strategy (replicating portfolio)

• By a change of variables, Black Scholes eqⁿ transforms into heat eqⁿ

$$\frac{\partial u}{\partial z} = \frac{\partial^2 u}{\partial x^2}$$

Boundary & Final Conditions $\Rightarrow$

For European Call

$$C(S, T) = \max(S - K, 0) \\ C(0, t) = 0, \quad C(\infty, t) \cong S$$

For European Put

$$P(S, T) = \max(K - S, 0) \\ P(0, t) = K e^{-r(T-t)}, \quad P(\infty, t) = 0$$

For European Call/Put E_t^Q can be solved analytically

$$C_t = e^{-r(T-t)} \left(e^{r(T-t)} S N(d_1) - K N(d_2) \right)$$

$$P_t = e^{-r(T-t)} \left(K N(-d_2) - e^{r(T-t)} S N(-d_1) \right)$$

where,

$$d_1 = \frac{\ln(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma \sqrt{T-t}}$$

$$d_2 = \frac{\ln(S/K) + \left(r - \sigma^2/2\right)(T-t)}{\sigma \sqrt{T-t}}$$

$N(x) \equiv$ Cumulative distributive function

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-u^2/2} du$$

$$f_t = e^{-r(T-t)} E_Q[f_T]$$

Q is risk neutral measure under which

$$dS = r S dt + \sigma S dW$$

PDF $\equiv$ Probability Density Function

$$\text{PDF}(S_T) = \frac{1}{\sigma S \sqrt{2\pi T}} \exp \left(- \frac{\left(\ln(S_T/S_t) - (r - \sigma^2/2)(T-t) \right)^2}{2\sigma^2(T-t)} \right)$$



$\Rightarrow$



$\Downarrow$



$$\Rightarrow C_T - P_T - S_T = -K$$

$$C_t - P_t - S_t = -K e^{-rt}$$

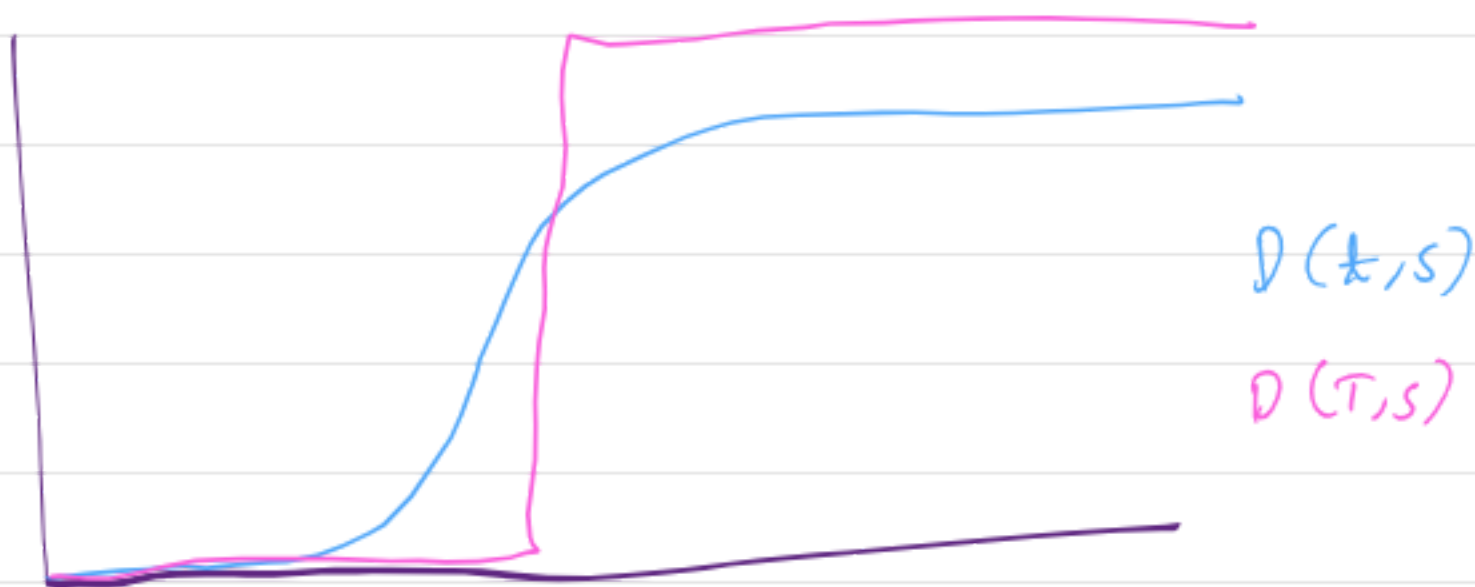
$$P = S - K e^{-rt} = C$$

Digital Contract.

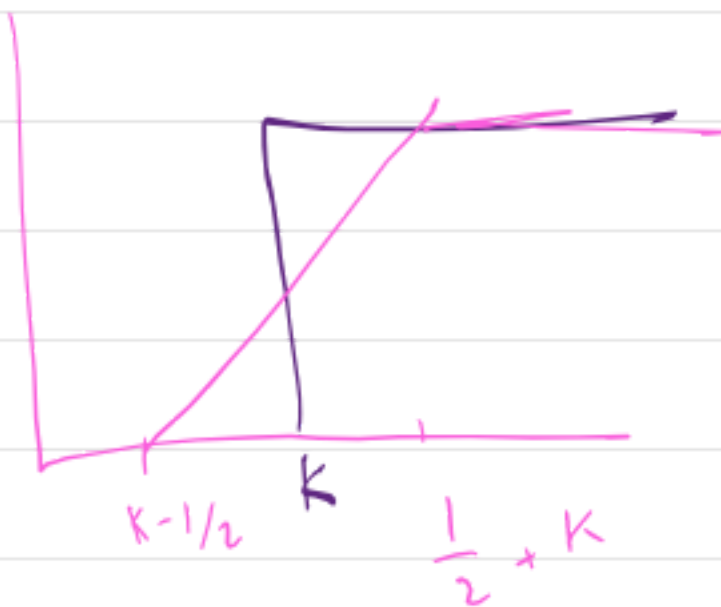
Strikeout K

Fixed Payoff B

$$\Rightarrow \text{If } \begin{cases} S > K \\ S \leq K \end{cases} \quad \begin{array}{l} \text{Payoff} \\ B \\ 0 \end{array}$$



$\Rightarrow$



To remove slope

$$\frac{C(K - \varepsilon) - C(K + \varepsilon)}{2\varepsilon}$$